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Nearshore
Offshore
Water temperature
Climate change
Water intake
Quantile regression

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Due to the interaction of wind-driven mixing and transport with basin boundaries, gravity and the rotation of the earth, complex patterns of thermal structure are found in the Great Lakes (Huang et al., 2010; Rao and Schwab, 2007; Rao and Sheng, 2008). In the spring, nearshore surface waters warm up sooner than offshore surface waters resulting in the formation of a thermal bar at 4 °C separating coastal and offshore waters (Mortimer, 2006). In the summer the lakes stratify, permitting intermittent wind events to generate horizontal gradients in surface temperatures as warmer, lighter waters are pushed downwind and cooler, heavier waters upwell upwind (Csanady, 1977; Simons and Schertzer, 1987). In the fall, inshore and offshore waters can become thermally distinct again as shallower, inshore waters cool more rapidly than deeper offshore waters (Chubarenko and Hutter, 2005). Water intakes for municipal and other uses are typically sited at various depths in the nearshore zone or in the inshore portion of the coastal boundary layer, and they can be differentially affected by the complex of water motions.

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Besides a range of water quality measurements to meet safety and environmental regulations (Edsall and Charlton, 1997), water temperature has been monitored daily often for decades at many municipal water intakes, for example, the data records examined by McCormick and Fahnenstiel (1999) for evidence of climatic trends. While the temperatures seen at shallower intakes typically track nearshore surface conditions, temperatures at deeper intakes also show evidence of the intermittent effects of upwelling (Trumpickas et al., 2009), similar to the surface patterns reported by Howell et al. (2012) at sites in the nearshore of Lake Ontario when upwelling events penetrated to the lake surface. Howell et al.'s (2012) observations also exhibit both advanced warming of nearshore surface waters in spring and advanced cooling in fall, relative to lake-wide mean surface temperatures, provided by the National Oceanic and Atmospheric Administration's Great Lakes Surface Environment Analysis (GLSEA) program (<http://coastwatch.glerl.noaa.gov/statistic/statistic.html>) (Schwab et al., 1999). While the GLSEA provides an estimate of lake-wide mean surface temperatures, the estimates largely reflect offshore conditions because (i) >75% of the area of each Great Lake is over depths >20 m (with the exception of Lake Erie, with ~41% of its area over depths of >20 m) (based on lake morphometry data from the Great Lakes Information Network <http://coastwatch.glerl.noaa.gov/statistic/physical.html>) and (ii) GLSEA pixels (taken on a 2.5 km scale) overlapping the shoreline are removed from the GLSEA water temperature estimate, further decreasing the influence of nearshore areas on the estimate of the overall lake-wide mean surface water temperature (Schwab et al., 1999). Furthermore, Schwab et al. (1999) showed high similarity between in-situ water temperatures recorded at eight offshore buoys across the Great Lakes and GLSEA temperatures measured at the buoy locations (mean temperature differences between buoy and GLSEA temperatures were <0.5 °C for all buoys), showing that the GLSEA program accurately measures offshore temperatures, but its relevance for nearshore temperatures is unknown.

Trumpickas et al. (2009) used empirical models linking air temperatures to GLSEA temperatures to project how climate change might affect future Great Lakes lake-wide surface water temperatures. These water temperature projections have proved useful in assessing possible impacts of climate change on Great Lakes' fish ecology and fisheries management (Lynch et al., 2010) and on regional biodiversity (Hall, 2012). In exploratory analyses, Trumpickas et al. (2009) showed that there were marked differences in seasonal water temperature patterns among specific nearshore sites. They hypothesized that these among-site differences were related to site-specific characteristics which affect both seasonal warming/cooling patterns and the influence of offshore hypolimnetic water (Trumpickas et al., 2009). These hypotheses are supported by recent empirical findings that: (i) link nearshore water temperature patterns to site bathymetry and offshore temperatures (Murphy et al., 2011; Wells and Parker, 2010) and (ii) demonstrate temporally consistent (i.e. over 36 years) statistical associations between Lake Superior offshore water temperatures and a nearshore site located in the lake's outflow (Austin and Colman, 2008).

Given the importance of Great Lakes' nearshore waters, there is a need for quantitative descriptions of how nearshore water temperature variability is shaped by nearshore site characteristics and offshore water temperatures. Given the complex physics that must be considered in attempting to mechanistically model the hydrodynamics of nearshore systems (Monismith et al., 1990; Rao and Schwab, 2007; Wells and Sherman, 2001), a formal assessment of the character and strength of the statistical associations linking site-specific seasonal patterns of water temperature variability to local characteristics of nearshore sites would be of value because: (i) it would define constraints on the historical behavior of these systems that would need to be accounted for by future mechanistic modeling and (ii) it would provide simple empirical tools for scoping the possible impacts of increasing surface temperatures due to climate change on nearshore Great Lakes water temperatures — tools that would complement

what Trumpickas et al. (2009) developed for lake-wide water temperatures.

The objectives of this paper are to use observed records of daily water temperatures from 28 Great Lakes nearshore sites to: (i) develop a single index that characterizes, for each site, the pattern of day-to-day changes in the historical distribution of nearshore–lake-wide water temperature differences over the period of positive stratification (when surface water temperatures are >4 °C), (ii) provide a quantitative description of how much of the site to site variation in this index is related to easily measured site characteristics like depth and fetch and (iii) use results from steps (i) and (ii) to develop a tool for scoping some of the potential impacts of increasing surface temperatures on midsummer water temperature patterns at nearshore sites around the Great Lakes. This approach will help identify primary drivers of nearshore temperature variation at the basin scale, provide initial forecasts of future nearshore temperature conditions, and identify promising approaches for refining next-generation forecasts.

Methods

Characterizing site-specific seasonal patterns of temperature variation

Data sources and sites

Daily nearshore water temperatures were compiled from 28 water intake sites from across the Great Lakes with 4 to 58 years of data per site (Fig. 1, Electronic supplementary material (ESM) Table S1). Intake data associated with each site include, at a minimum, daily water temperature, the depth of the water intake (the depth at which the temperature was recorded), and a geographical location for the water intake. Some of these data were also used by McCormick and Fahnenstiel (1999) to analyze Great Lakes nearshore temperature trends, but most of the data used here were gathered through direct contact with staff at individual water filtration plants. Each Great Lake was represented by between four and nine intake sites. The earliest water temperature record was from 1943 and the latest from 2000. The 28 intake sites cover much of the range of fetch, exposure and depth values present in the nearshore zone of the Great Lakes — therefore, these sites were considered a representative sample of the nearshore zone of this interconnected set of water bodies. Analyses described in the following sections were completed for all sites across lakes together rather than completed separately for each lake because we expected nearshore water temperature variation to be influenced more by site-level characteristics than lake-level characteristics. This expectation is based on the knowledge that, for all the lakes: (i) thermocline depths are generally similar in stratified regions of the lakes (Boyce et al., 1989; Mazumder and Taylor, 1994); (ii) hypolimnetic temperatures are similar (<http://www.glerl.noaa.gov/res/glcfs/>); and (iii) prevailing winds are similar (Boyce et al., 1989).

Trumpickas et al. (2009) developed empirical models for lakes Superior, Huron, Erie and Ontario that were designed to estimate GLSEA-based lake-wide daily mean surface water temperatures from daily regional air temperatures for the period of positive stratification (>4 °C). We used the same approach to derive a similar model for Lake Michigan, based on GLSEA data for that lake for the years from 1995 through 2006. We then used these five lake-specific models and local air temperatures for lakes Superior, Huron, Erie, Ontario and Michigan (source: Environment Canada weather stations at Sault Ste. Marie, Sudbury, Windsor and Trenton, <http://www.climate.weatheroffice.ec.gc.ca> and National Oceanic and Atmospheric Administration National Weather Service weather station in Manitowoc, Wisconsin, compiled by the Goddard Institute for Space Studies http://data.giss.nasa.gov/gistemp/station_data/, respectively), to estimate GLSEA-based lake-wide daily mean surface water temperatures for each lake for all the years for which we had intake data. We had to rely on these indirect estimates of past lake-wide surface water temperatures, rather than direct GLSEA estimates, because the bulk of the data from most of our

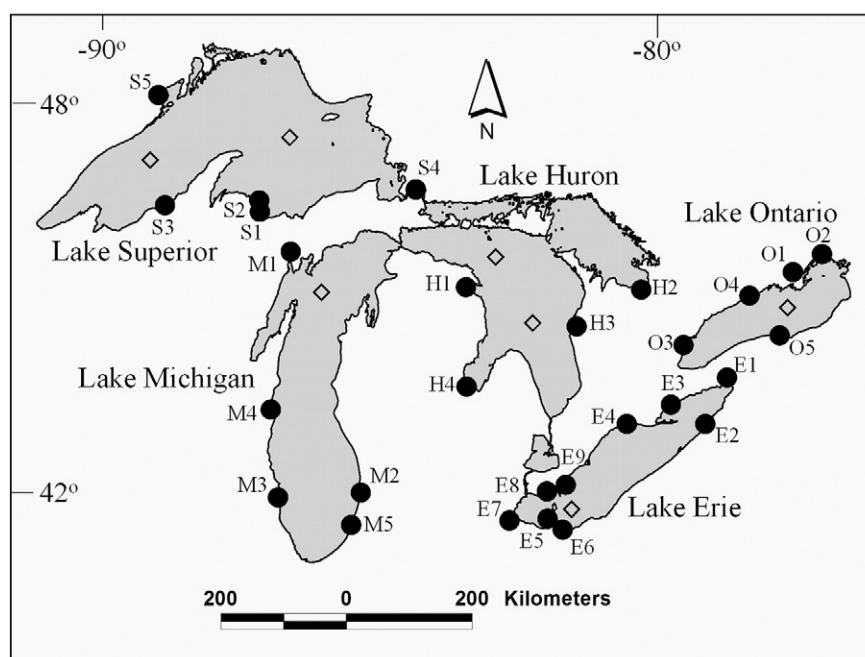


Fig. 1. Locations of Great Lakes water intake stations (solid circles) and wind buoys (hollow diamonds) used in this study. Labels for water intake stations correspond with ESM Table S1.

sites came from years prior to the start (1995) of the GLSEA remote-sensing system. We assessed the accuracy of these indirect estimates by comparing them with observed GLSEA values for six years of data from each of the five lakes where observed GLSEA values were available and found that 87% of the yearly averaged differences between observed and estimated daily temperatures were less than ± 0.5 °C and were not consistently positive or negative. In addition, these estimates of daily lake-wide mean surface temperatures were validated against independently collected offshore surface water temperatures for the period 1968–2002 (Trumpickas et al., 2009), demonstrating that (i) these estimates of surface water temperatures are reliable and (ii) lake-wide mean surface temperatures are very similar to offshore surface water temperatures.

Simulating expected seasonal patterns of variation

In characterizing nearshore temperature variation, we felt that it was important to explicitly recognize the strong role that offshore conditions play in shaping the thermal environments found at nearshore intake sites on very large lakes. Therefore we focused our study on the daily difference between lake-wide mean surface water temperatures (which are reflective of offshore conditions) and site-specific nearshore water temperatures. This focus also established a direct link between our analysis of nearshore conditions and the lake-wide temperature models developed in our previous study (Trumpickas et al., 2009). We used the semi-mechanistic seasonal temperature-profile model (STM) of Minns and Shuter (2013) (see ESM Appendix S1 for a full explanation of the simulation methods) to illustrate the changes in the shape of the seasonal pattern of temperature variation likely to be observed with changes in depth at a 'typical' Great Lakes nearshore site. Our objective with the semi-mechanistic model was to simulate interactions among the primary nearshore physical phenomena (e.g. seasonal appearance/disappearance of stratification, sporadic occurrence of seiche activity). This simulation helps show how observed empirical patterns in the nearshore temperature data are directly related to parameters in the STM (i.e. depth, thermocline movement, differential nearshore/offshore warming, etc.). First, we parameterized the STM to represent the offshore region of a large lake. These parameters were then used as baselines to set up a second version of the STM that would capture known relationships between nearshore and offshore

temperatures (i.e. faster nearshore warming in the spring and cooling in the fall), and also incorporate upwelling-downwelling effects via vertical movements of the thermocline. We assumed: (i) significant (± 5 °C) and random inter-annual variation in maximum summer temperatures (both nearshore and offshore), (ii) a common mean thermocline depth nearshore and offshore, and (iii) significant (± 5 m), random daily variation in thermocline depth nearshore, reflecting upwelling/downwelling events. We then generated 20 years of nearshore and offshore temperature data and summarized the seasonal pattern of nearshore temperature variation at each depth by plotting the daily distribution of temperature differences between offshore surface values and nearshore depth-specific values for each day from spring to fall. We then compared these seasonal patterns of variation across depth and found changes in shape (Figs. 2a,b) that matched the differences in shape in the observed nearshore water temperature data (Figs. 2c,d). Although the STM simulations rely on offshore temperatures and our empirical analyses rely on lake-wide mean temperatures (see descriptions above), offshore and lake-wide mean temperatures are quite similar: (i) estimates of daily lake-wide mean surface temperatures were similar to independently collected offshore surface water temperatures (Trumpickas et al., 2009) and (ii) GLSEA lake-wide mean temperatures compared favorably to offshore buoy temperatures (Schwab et al., 1999).

Characterizing observed site-specific seasonal patterns of variation

For each intake site, daily temperature differences between nearshore water intake readings and lake-wide (GLSEA) surface water temperature estimates were pooled across years, plotted against day of the year and the resulting seasonal pattern of variation in temperature differences compared to the patterns generated in our simulation study. The observed patterns exhibited a range of shapes (examples of patterns shown in Figs. 2c,d) that closely matched the range produced in our simulation study: for shallow sites, the width of the daily distribution of temperature differences was typically fixed and independent of day of year (Figs. 2a,c); for deeper sites, the width of the distribution often grew rapidly from spring to mid-summer and then shrank progressively from mid-summer through fall (Figs. 2b,d).

The primary difference between these patterns lies in the presence/absence of systematic seasonal changes in the boundaries of the daily

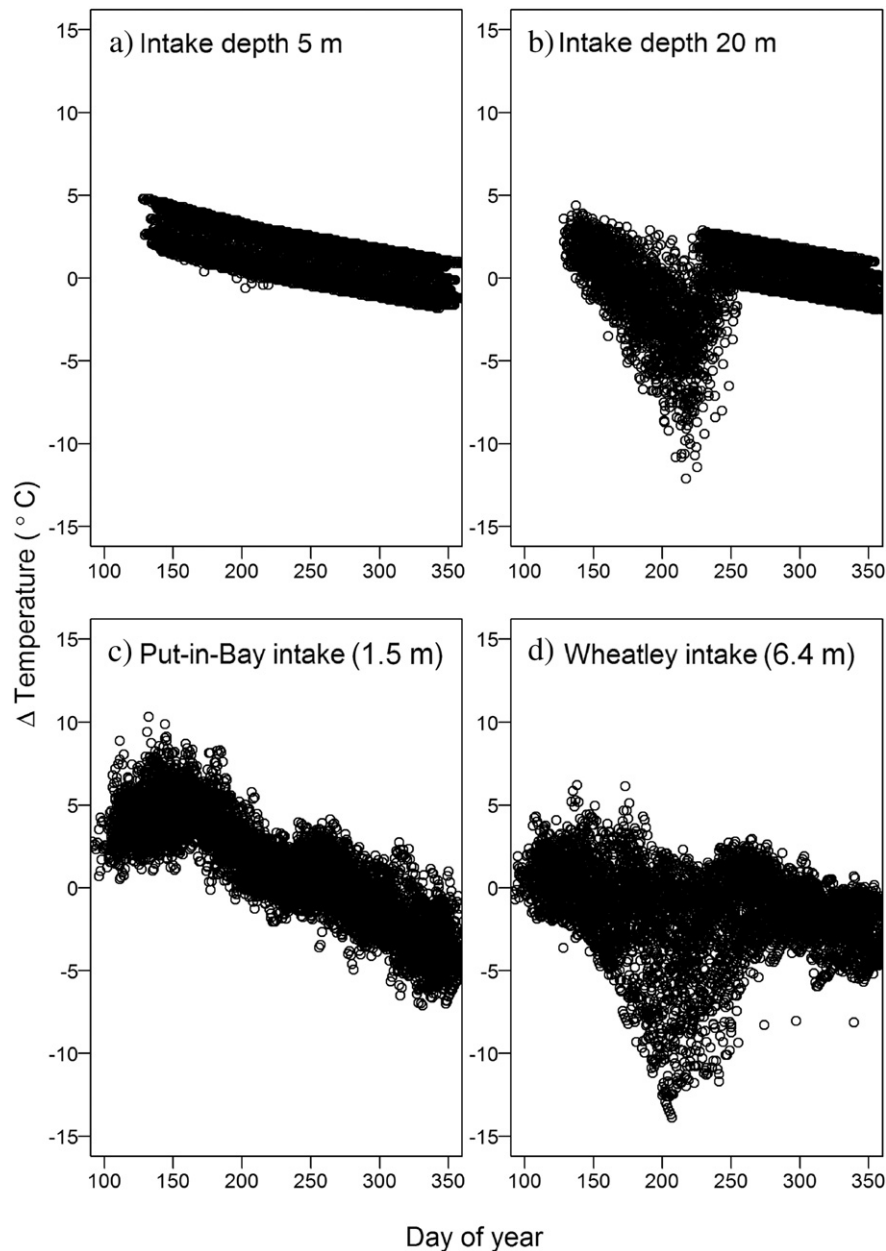


Fig. 2. Simulated and observed temperature differences between nearshore water intakes and lake-wide surface temperatures. The range of simulated seasonal patterns observed across all depths is illustrated by the results at (a) 5 m and (b) 20 m depth. The simulations were parameterized using Lake Ontario values, however, results were typical of observed patterns across all the Great Lakes. Similar seasonal patterns were observed at many nearshore sites and are illustrated by two nearshore sites on Lake Erie: (c) Put-in-Bay (site E5 in Fig. 1; depth 1.5 m) and (d) Wheatley (E9; depth 6.4 m). Panels (a) and (c) illustrate the pattern of a constant residual spread from spring to fall with warmer nearshore waters in spring and cooler in fall. Panels (b) and (d) illustrate the pattern of the residual spread widening and contracting from spring to fall, with the largest negative residuals in midsummer.

distribution of temperature differences. Quantile regression is particularly suited to empirical studies of such effects since it is designed specifically to identify consistent associations between distributional benchmarks (e.g. the 5% quantile, the 95% quantile) and a set of independent variables. It does this by fitting simple functions of the independent variables to each quantile of the distribution (Cade and Noon, 2003).

We used quantile regression to generate a site-specific measure of the seasonal changes in the daily distribution of nearshore–lake-wide temperature differences. For each site, we used the quantreg (Koenker, 2009) and splines packages in R 2.10 (R Development Core Team, 2010) to fit 2nd order models (i.e. $q_i = a_i + \beta_i d + \delta_i d^2$, where d is day of year, i is quantile value, q_i is the i th quantile of the daily distribution of nearshore–lake-wide lake temperature difference and $[a_i, \beta_i,$

$\delta_i]$ are the fitted model parameters for the i th quantile) to the seasonal changes in the quantiles (i.e. every 0.01 quantile from 0.02 to 0.98) of the daily distribution of nearshore–lake-wide temperature differences. The quadratic model provided a better fit (based on AICc values) than the null (intercept-only) model for each site. The coefficient of the 2nd order term (δ) for each 0.01 quantile from 0.02 to 0.98 was then least-squares regressed against the quantile value (Fig. 3). The slope (the Seasonal Variation Index – SVI) from this regression provides a synoptic measure of seasonal change in the distribution of daily nearshore–lake-wide temperature differences that captures the range of expected seasonal changes; the expectation was that (i) an SVI value near zero is indicative of a distribution whose width does not change over the open-water season (Figs. 2a,c) and (ii) a negative SVI value is

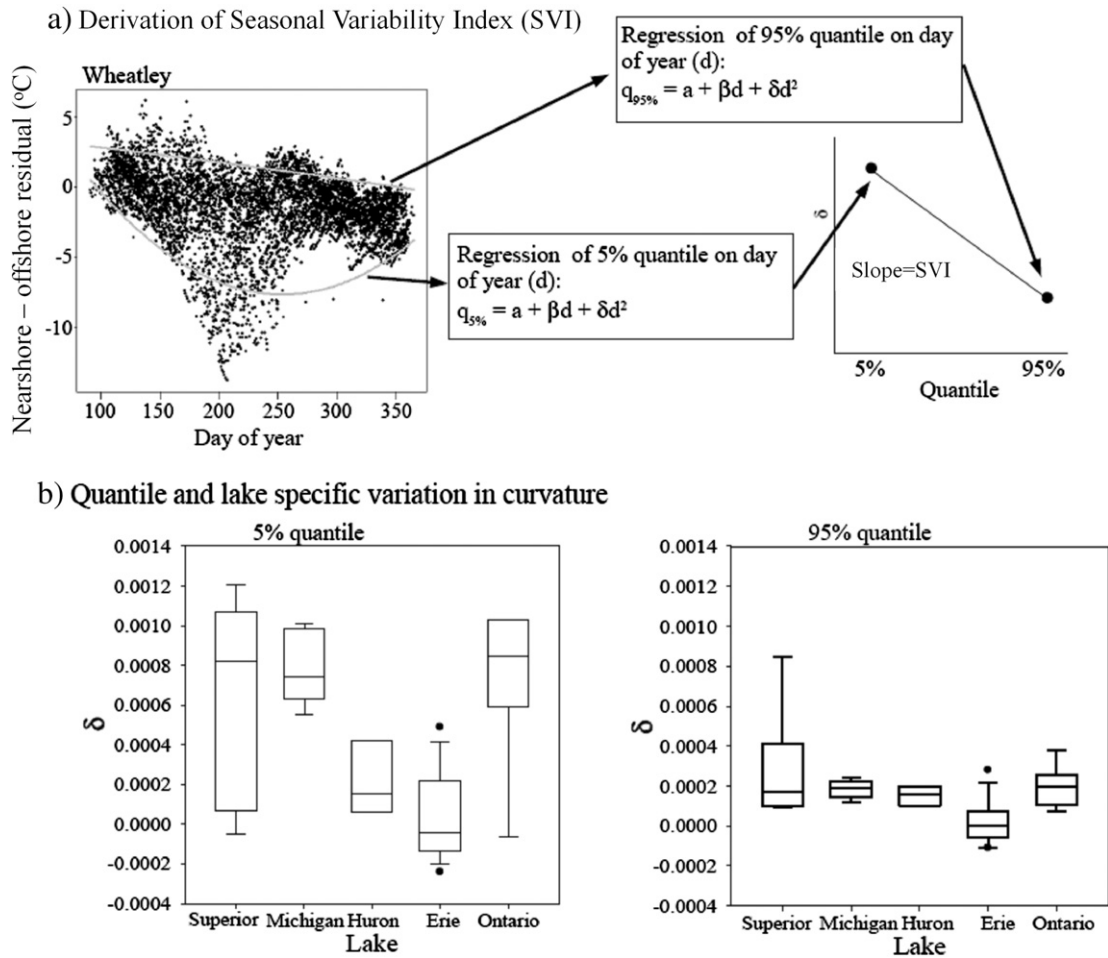


Fig. 3. (a) Schematic diagram demonstrating the derivation of the Seasonal Variability Index (SVI) and (b) box plots showing the lake specific variation in curvature (δ) of the 5% and 95% quantiles. Lower and upper bounds on boxplot rectangles represent the 25th and 75th percentiles, respectively, while dots represent values less than or greater than the 10th and 90th percentiles, respectively.

indicative of a distribution whose width waxes and then wanes over the open-water season (Figs. 2b,d).

Factors driving seasonal patterns of temperature variation

Our objective was to identify local site variables that are associated with different seasonal nearshore temperature patterns, and therefore can be used as simple indicators for nearshore temperature variability patterns at different sites on the Great Lakes. The objective was not a mechanistic explanation of the detailed shifts in observed patterns of variability, but a simple empirical description of the broad, site-to-site differences in patterns that are observed to occur. Thus, we assessed the strength of statistical associations linking SVI values to the site characteristics that we hypothesized would be related to it. First, we investigated intake depth, mean fetch and maximum fetch. Intake depth was provided by each water intake utility. We expected that sites at greater depths would be colder, relative to lake-wide surface water temperatures, than shallower sites, as well as more likely to be influenced by upwellings of hypolimnetic waters (Wells and Parker, 2010). In addition, simulation results (see above) showed intake depth to be a key factor in differences between seasonal temperature patterns. Intake depth was considered constant at a site over our study period because typical inter-annual variation in Great Lakes water level is only 40–50 cm (Lenters, 2001). Mean fetch was calculated as the mean distance to shore (in meters) from the site across 16 compass directions (equally spaced compass directions starting at

north). Maximum fetch was the largest of these 16 values. Both mean and maximum fetch are measures of the ‘openness’ of the site to the main body of the lake. Therefore, we expected sites with high mean (or maximum) fetch to be more likely to be affected by upwellings and downwellings than sites with low fetch. Furthermore, sites with low fetch tend to be located in embayments that warm and cool faster than the main body of the lake to which they are attached (Murphy et al., 2011).

We then introduced a seasonal time-dependent element to the fetch value for each site by developing an average exposure index (E_M) for each month (M) as follows: (i) for each month and year, we calculated $\sum w_i p_i f_i$, where w_i is the mean wind speed (meters/second) blowing from direction i for the month and year, p_i is the proportion of time the wind blew from direction i for the month and year, and f_i is the fetch in direction i (Keddy, 1982); (ii) for each month, we then averaged these values across years to get E_M . Sixteen compass directions were used to calculate exposure. Wind data were provided by National Data Buoy Centre buoys located in each of the lakes (two buoys each in Lakes Superior and Huron; one buoy each in Lakes Michigan, Erie and Ontario; Fig. 1; a second buoy is present in Lake Michigan but it has not been collecting continuous wind data for the time period of interest). Site exposure was calculated using wind data from the nearest buoy to each site (hence excluding a third buoy that is present on Lake Superior). Exposure for each site was calculated for the months of May to November for the years 2004–2009. The wind data used in our exposure estimates did not overlap temporally with the water

temperature data because the Lake Ontario buoy only began operation (2004) after the end of the time period covered (1943–2000) by our water temperature dataset. In some cases, the buoys used for wind data are several hundred kilometers from our nearshore sites; these data were still used because (i) we were interested in assessing the role of regional-scale prevailing wind conditions, not specific local conditions and (ii) we calculated average monthly exposure values for sites on Lakes Huron and Superior (i.e. those lakes with continuous wind data recorded for two buoys from 2004 to 2009) using wind data from the further away buoy and found that these exposure values were highly correlated with exposure values calculated using the closer buoy (Huron: $r^2 = 0.82$; Superior: $r^2 = 0.90$), showing that even distant wind observations are useful in determining broad-scale wind conditions. Our goals were to assess whether a simple measure of wind patterns from six years of buoy data would be sufficient to derive a 'typical' exposure value for each site that would be useful in explaining variation in SVI values and whether these simple wind variables would add explanatory power beyond the depth and fetch terms already included. Exposure was expected to influence nearshore water temperatures in a similar way as fetch with high exposure sites more likely to be affected by upwellings and downwellings than low exposure sites. Because exposure incorporates both fetch and wind effects and both are important factors in determining upwelling and downwelling events (Boyce et al., 1989), in principle, we expected wind-based exposure variables to be better predictors of SVI than fetch variables alone.

Finally, we also examined a lake- and time-dependent variable: $T_{\max}$ — a lake-specific measure of the peak daily lake-wide surface water temperature for a year (see Trumpickas et al., 2009 for definition and derivation of predictive models for $T_{\max}$). We estimated $T_{\max}$ values for each year and lake in our nearshore water temperature database and included these estimates in our regression models of SVI to account for (i) lake-specific and (ii) year-specific differences in the annual lake-wide temperature cycle. We expected such differences to affect the magnitude of the nearshore–lake-wide temperature difference because surface waters are expected to warm more quickly than hypolimnetic waters under climate change (Lehman, 2002), thereby amplifying the nearshore cooling caused by hypolimnetic upwellings. Another variable frequently linked to water temperature variation (i.e. air temperature) was already explicitly included in the Trumpickas et al. (2009) model used to generate lake-wide mean surface water temperatures (see above) and therefore was not explicitly included in our statistical analyses of observed intake-site water temperatures. However, it was considered explicitly in our projections of future surface temperatures (see below).

We used linear least-squares regression to examine the relationships between SVI and the following independent variables: intake depth, mean fetch, maximum fetch, monthly exposures and $T_{\max}$. First, we examined bivariate relationships between SVI and the independent variables to test whether site characteristics affected nearshore water temperature patterns as expected. Next, we used best subsets regression with Aikake's Information Criteria corrected for small sample size (AICc) to pick the 'best' model capable of accounting for variation in SVI from two pools of models. The first pool of models we considered included depth, monthly exposures and $T_{\max}$ as candidate variables. The second pool of models included depth, mean fetch, maximum fetch and $T_{\max}$ as candidate variables. All interaction terms between $T_{\max}$ and the other variables were included in the best subsets analysis for both pools of models.

Projecting impacts of changes in surface temperatures on nearshore temperature variability

Typical climate change scenarios for the Great Lakes region (e.g. the IPCC A2 CO₂ emission scenario; Nakićenović et al., 2000) project long term increases in summer air temperatures. Trumpickas et al. (2009)

showed that such changes would likely translate into increases in lake-wide surface water temperatures. To examine the effects of increasing surface water temperatures due to climate change on nearshore temperature variability, we focused on the midsummer period — the period of the year where we expected the greatest variability in nearshore, depth-specific water temperatures, relative to lake-wide mean surface water temperatures. We expected high variability in midsummer because surface temperatures are at their maximum at this time of year (Trumpickas et al., 2009) and thus the impact of hypolimnetic upwellings on nearshore temperatures will also be at its maximum (Wells and Parker, 2010). We chose the day of year 225 (13 August) as being representative of the midsummer period; a single day rather than a range of days was used for simplicity and to represent a single common peak in nearshore–lake-wide temperature differences. Day 225 in particular was chosen to characterize midsummer because this day had the greatest observed range of nearshore–lake-wide temperature differences across all sites. We used three 'benchmark' values (the 5%, 50%, 95% quantiles) to characterize the spread of nearshore–lake-wide temperature differences at midsummer for each site and, for consistency across sites, we estimated them as follows: (i) 2nd order quantile regression models were used to quantify daily changes in each quantile and (ii) these models were then used to estimate the value of each quantile on day 225, thus generating the 5%, 50% and 95% benchmarks, respectively. We compared the value of each quantile on day 225 to the average value of quantiles for the 30 days surrounding day 225 (day 211 to day 240) and found single-day and 30-day values to be very similar for all quantiles and sites (5% quantile: mean difference between day 225 benchmark value and days 211–240 averaged benchmark values = 0.022 °C, standard deviation = 0.036; 50% quantile: mean difference = 0.0057 °C, standard deviation = 0.023; 95% quantile = −0.0015 °C, standard deviation = 0.016). Thus, day 225 is representative of the midsummer time period when nearshore–lake-wide water temperatures show the greatest divergence.

Our objective was to characterize each of our 28 sites with these benchmark values and then develop simple empirical models of the benchmarks to project how increasing surface temperatures might affect them. These projections would set reasonable bounds for the likely effects of increased surface temperatures on the range of midsummer water temperatures found in the nearshore of the Great Lakes. Our approach assumes that hypolimnetic temperatures would remain constant in the future and only takes into account increased surface temperatures, not other factors affected by climate change. Our approach was as follows: (i) we used linear regression to develop models for predicting values of the 5%, 50% and 95% benchmarks from the subset of site-specific characteristics (intake depth, mean and maximum fetch and monthly exposure) and lake-specific ($T_{\max}$) characteristics that were associated with SVI, (ii) we generated projections (Trumpickas et al., 2009) for lake specific $T_{\max}$ values under the IPCC A2 CO₂ emission scenario (Nakićenović et al., 2000) and used them in our benchmark models to project future benchmark values for specific sites and (iii) we then used projections (Trumpickas et al., 2009) of whole lake surface water temperatures for day 225 under the IPCC A2 CO₂ emission scenario to convert site-specific projections of our three benchmarks into a site-specific projection of the nearshore water temperature distribution on day 225. Linear least-squares best subsets regression with AICc was used to choose the best combination of independent variables to predict each benchmark (5%, 50% and 95%). Two analyses were performed. One analysis used monthly exposure values with intake depth, $T_{\max}$ and interaction terms, and the other used mean and maximum fetch instead of exposure terms (again with intake depth, $T_{\max}$ and interaction terms as the other candidate variables).

We chose four representative sites to illustrate our projections of how future increases in air temperature could affect Great Lakes' nearshore water temperatures. The four sites were purposely chosen to represent the range of fetch and depth values in the dataset and were

chosen from four different lakes: Marquette–Superior (S1 in Fig. 1; deep and high fetch), Kincardine–Huron (H3; shallow and high fetch), Sandusky–Erie (E6; shallow and low fetch), and Kingston–Ontario (O2; deep and low fetch) (see ESM Table 1 for depth and fetch values). The climate change scenario chosen, the IPCC A2 CO₂ emission scenario, assumes a rapid increase in world human population to 15 billion by 2100 with a 3.5 times increase in annual global CO₂ emissions by 2100 compared to 2000 levels (Nakićenović et al., 2000). The A2 scenario assumes higher emissions than other IPCC scenarios; therefore, our projections likely suggest greater changes in nearshore water temperatures than other scenarios might. Climatic changes for this scenario were generated by the Canadian Global Climate Model Version 2 (CGCM2) (Flato et al., 2000; Flato and Boer, 2001). Projections of $T_{\max}$ for each Great Lake under the A2 scenario were obtained from Trumpickas et al. (2009) for three time periods: 2011–2040, 2041–2070 and 2071–2100. These $T_{\max}$ values were used in the regression model for each benchmark to generate projections of benchmark values for each of our four sites. Because benchmark values characterize the midsummer distribution of nearshore–lake-wide water temperature differences, we used projections of lake-wide surface water temperatures (from Trumpickas et al., 2009) to obtain the actual predicted nearshore temperatures associated with each benchmark for each of the four representative sites.

Results

Characterizing site-specific seasonal patterns of temperature variation

Nearshore–lake-wide water temperature differences plotted against day of year from the 28 sites exhibited two predominant patterns (Figs. 2c,d). The first pattern (Fig. 2c) was characterized by relatively constant variance throughout the year, with positive values (nearshore temperatures warmer than lake-wide surface temperatures) predominating early in the year, followed by a steady decline as the year progressed, with mostly negative values observed at the end of the year. Within this pattern, there were site-specific differences in the amount of variation observed throughout the year, the range of positive values observed early in the year and the range of negative values observed late in the year. The second pattern (Fig. 2d) resembled a 'V-shape' and was characterized by an expansion followed by a contraction of variance as day of year increased. The lower portion of the distribution in midsummer was negative, reflecting nearshore subsurface temperatures that were cooler than the lake-wide surface temperature. Within this pattern, there was variation across sites in the values of the maximum negative temperature difference observed in midsummer, as well as the values of the maximum positive differences observed throughout the year. As expected, our seasonality index (SVI) captured the observed site-to-site differences in these patterns (Fig. 3): (i) a very negative value for the SVI ($SVI < -2.4 \times 10^{-4}$) identified a pattern where the spread of the distribution of temperature differences peaked in midsummer (defined here as day of year 195 (July 14) to 255 (Sept 12)), with $\geq 5\%$ of midsummer temperature differences $< -5^\circ\text{C}$, (ii) a less negative value ($-2.4 \times 10^{-4} < SVI < 0$) for the SVI identified a pattern where the spread of the distribution of temperature differences peaked in midsummer but with fewer ($\leq 5\%$) midsummer temperature differences $< -5^\circ\text{C}$ and (iii) an SVI value greater than or equal to 0 identified a pattern where the spread of residuals was relatively constant throughout the year. Values of SVI ranged from -9.2×10^{-4} (for Holland on Lake Michigan) to $+2.4 \times 10^{-4}$ (for Saginaw on Lake Huron) (ESM Table S1).

Factors driving seasonal patterns of temperature variation

Almost all of the hypothesized linear relationships between SVI and site- and lake-specific factors were evident in the observed data. There were negative relationships (one sided $p < 0.005$) with both intake

depth and mean fetch (Fig. 4), exposure relationships from May through October were negative and all but the July relationship were significant (for all months but July: one-sided $p < 0.03$; July one-sided $p < 0.075$). The only variables not associated with SVI were maximum fetch, November exposure and maximum summer temperature ($T_{\max}$). In all of the significant relationships, there was no obvious tendency for sites from the same lake to group together (Fig. 4).

Best subsets regression modeling with depth, monthly exposures and $T_{\max}$ as candidate variables showed that 29 models predicting SVI had substantial support (i.e. $\Delta\text{AICc} < 2$; Burnham and Anderson, 2002). Of these, we picked two 'preferred' models based on the following criteria: (i) both had low ΔAICc values, (ii) both were relatively simple, including a relatively low number of independent variables and (iii) both contained exposure terms from months (June and September) that are known to be within the upwelling season of the Great Lakes. Of the two models selected (Table 1), model 1 containing the interaction term, is intuitively appealing because the addition of $T_{\max}$ accounts for lake- and year-specific differences in whole lake surface temperatures which should directly affect the nearshore–whole lake temperature residuals (and therefore SVI). Alternatively, model 2, with no interaction terms, is simpler and is indistinguishable from model 1 in its predictive power. Overall, both models can explain a large proportion ($r^2 = 0.75$) of the observed variability in SVI.

When considering depth, mean and maximum fetch and $T_{\max}$ as candidate variables, two models were preferred from those with $\Delta\text{AICc} < 2$ (Table 1). Model 3 ($r^2 = 0.73$) is simpler, with only mean and maximum fetch and intake depth as independent variables. Model 4 contains interaction terms with $T_{\max}$, and is, as above, appealing because the addition of $T_{\max}$ accounts for lake- and year-specific differences in lake-wide surface temperatures, and has a slightly higher r^2 value ($r^2 = 0.74$).

The high degree of predictability in SVI, plus the absence of lake-specific clustering of residuals in all regression models, supports our hypothesis that the range of likely temperatures at a site is shaped to a significant degree by lake-independent local variables like site depth and fetch.

Projecting impacts of changes in surface temperatures on nearshore temperature variability

With intake depth, $T_{\max}$ and monthly exposure terms as candidate independent variables, four sets of models were identified that had ΔAICc values between two and four for each benchmark (the lowest ΔAICc values for models that were consistent across the three benchmarks). All four sets of models contained four terms consisting of combinations of $T_{\max} \times \text{depth}$, JulyExposure, AugustExposure, JulyExposure $\times T_{\max}$ and AugustExposure $\times T_{\max}$. The set with the best combination of low ΔAICc values and high r^2 values is reported in Table 1. This set was simpler than some of the other sets considered because it had two rather than three interaction terms.

With intake depth, $T_{\max}$ and fetch terms (mean and maximum fetch) as candidate independent variables, four sets of models (each containing four terms) were identified that had ΔAICc values < 2 for each of the benchmarks. The set with the best combination of low ΔAICc values, high r^2 values and fewer interaction terms is reported in Table 1. It was found that this set of fetch-based models had slightly higher r^2 values (and the same number of terms) than the preferred set of models using exposure terms. Therefore, the fetch-based set of models was used for projecting sensitivities to increased surface temperatures.

We generated projections of the midsummer range of nearshore water temperatures for four Great Lakes sites for the periods 1971–2000, 2011–2040, 2041–2070 and 2071–2100. Projections were made for the 5%, 50% and 95% benchmarks and these benchmarks were then converted to raw nearshore water temperatures (Fig. 5). Of the four sites, Kincardine was projected to have the least increase in nearshore water temperatures from 1971–2000 to 2071–2100, while

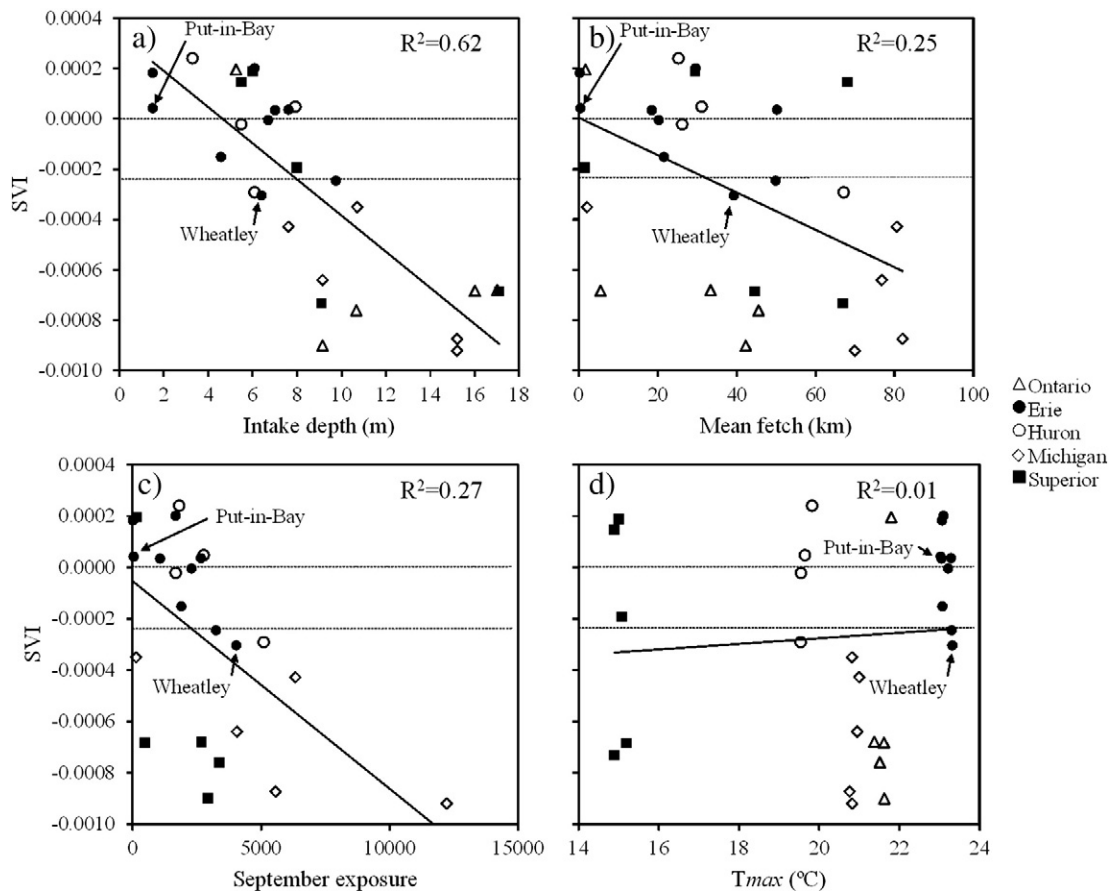


Fig. 4. Relationships between the Seasonal Variability Index (SVI) and (a) intake depth, (b) mean fetch, (c) September exposure (the monthly exposure with the strongest relationship to SVI), and (d) T_{max} . Symbols group sites by lake. The solid line in each plot represents the least-squares regression relationship linking SVI to depth, fetch, September exposure or T_{max} , respectively. The upper dashed horizontal line on each plot marks an SVI value of 0; sites with SVI values equal to/greater than this show an essentially constant residual spread from spring to fall. The lower dashed horizontal line marks an SVI value of -2.4×10^{-4} ; sites with SVI values less than this show a strong V-shaped pattern ($\geq 5\%$ of midsummer values $< -5^\circ\text{C}$). Sites in between the two lines show an intermediate seasonal pattern. Put-in-Bay (E5 in Fig. 1, Fig. 2c) and Wheatley (E9 in Fig. 1, Fig. 2d) are marked for reference.

Kingston was projected to have the greatest increase. At all sites, the nearshore temperature associated with the 95% benchmark was projected to increase the most over time and the nearshore temperature associated with the 5% benchmark the least, however these differences were relatively small.

Discussion

Seasonal changes in nearshore water temperature variation

Our empirical models for SVI (Table 1) can be used to scope the seasonal pattern of variation in water temperatures likely to be experienced at nearshore sites in the Great Lakes. In essence, the SVI value is a synoptic measure of the seasonal influence of cold hypolimnetic water on water temperatures observed at the site. The SVI prediction equations in Table 1, in particular the fetch-based Model 3, can be used to estimate the SVI for any nearshore site on the Great Lakes using easily measured values (fetch and depth). If a site's SVI value is $< -2.4 \times 10^{-4}$, it will exhibit a strong V-shaped seasonal pattern (e.g. Figs. 2b,d with $\geq 5\%$ of mid-summer values $< -5^\circ\text{C}$); if its SVI value is ≥ 0 , it will exhibit a band-like pattern of fixed width (e.g. Figs. 2a,c) and intermediate SVI values are associated with sites that exhibit patterns that are intermediate between these two extremes. For sites with low SVI values, the annual peak in temperature variation can be estimated using the summer quantile prediction equations specified in Table 1.

The patterns observed in our study mirrored findings from previous studies of nearshore water temperature patterns in lakes and our

hypotheses regarding relationships between the SVI and site characteristics were generally supported. For example, Wells and Parker (2010) found depth to be an important predictor of nearshore water temperature variability during the summer in Lake Huron, with depths of 10–20 m having more variability due to upwelling and downwelling events than shallower or deeper areas. This is similar to our finding that depth is strongly negatively related to SVI – indicating that deeper sites (maximum depth of our study sites was 17.1 m) showed a greater range of temperatures in the summer (probably due to upwellings and downwellings) than shallow sites. In addition, our simulation results with the seasonal temperature-profile model (STM) showed directly how changes in depth affect the seasonal nearshore temperature patterns characterized by SVI. As expected, we found sites with high mean fetch to have lower SVI values, suggesting that high fetch sites are more likely to be impacted by upwellings and downwellings. Furthermore, enclosed areas typically warm faster than the main lake (Bolgrien and Brooks, 1992; Murphy et al., 2011) as was indicated by our finding that low mean fetch was associated with sites with higher SVI values, and typically higher spring water temperatures than those found lake-wide (Fig. 2). Plattner et al. (2006) linked wind activities in Lake Michigan to nearshore upwellings, similar to our finding that exposure (which includes a measure of wind activity) was associated with lower SVI values. Although clearly more factors (considered below) than simply depth and fetch are needed to fully account for nearshore temperature characteristics, our findings do show that differences in fetch and depth are consistently associated with gross differences in seasonal patterns of temperature variation.

Table 1

Regression models for the Seasonal Variability Index (SVI) and midsummer benchmarks, with exposure terms and with fetch terms. For exposure-based SVI models $\Delta\text{AICc} < 0.7$ and $p < 0.0001$; for fetch-based SVI models $\Delta\text{AICc} < 1.04$ and $p < 0.0001$; for exposure-based benchmark models $\Delta\text{AICc} < 3.4$ and $p < 0.001$; for fetch-based benchmark models, $\Delta\text{AICc} < 2.0$ and $p < 0.001$.

Exposure-based models for Seasonal Variability Index (SVI)							
Model	Constant	Coefficients for independent variables				Statistics	
		Depth	June exposure	Sept exposure	$T_{\text{max}} \times \text{Sept exposure}$		
1	3.406×10^{-4}	-6.1×10^{-5}	1.425×10^{-7}	–	-6.969×10^{-9}	$R^2 = 0.75$; RMSE = 2.07×10^{-4}	
2	3.476×10^{-4}	-6.1×10^{-5}	1.276×10^{-7}	-1.356×10^{-7}	–	$R^2 = 0.75$; RMSE = 2.09×10^{-4}	
Fetch-based models for Seasonal Variability Index (SVI)							
Model	Constant	Coefficients for independent variables				Statistics	
		Depth	Mean fetch	Max fetch	$T_{\text{max}} \times \text{Mean fetch}$		$T_{\text{max}} \times \text{Max fetch}$
3	3.502×10^{-4}	-6.072×10^{-5}	-8.736×10^{-6}	1.215×10^{-6}	–	–	$R^2 = 0.73$; RMSE = 2.16×10^{-4}
4	3.524×10^{-4}	-5.984×10^{-5}	–	–	-4.453×10^{-7}	5.961×10^{-8}	$R^2 = 0.74$; RMSE = 2.12×10^{-4}
Exposure-based models for midsummer benchmarks of nearshore temperature distributions							
Benchmark	Constant	Coefficients for independent variables				Statistics	
		July exposure	August exposure	$T_{\text{max}} \times \text{July exposure}$	$T_{\text{max}} \times \text{depth}$		
5%	1.736	2.826×10^{-3}	-1.729×10^{-3}	-8.349×10^{-5}	–0.0280	$R^2 = 0.66$; RMSE = 2.594	
50%	3.446	3.888×10^{-3}	-7.64×10^{-4}	-1.53×10^{-4}	–0.0190	$R^2 = 0.61$; RMSE = 1.807	
95%	5.201	3.748×10^{-3}	-5.8×10^{-4}	-1.55×10^{-4}	–0.0109	$R^2 = 0.56$; RMSE = 1.447	
Fetch-based models for midsummer benchmarks of nearshore temperature distributions							
Benchmark	Constant	Coefficients for independent variables				Statistics	
		Depth	Mean Fetch	Maximum fetch	$T_{\text{max}} \times \text{Maximum fetch}$		
5%	0.796	–0.477	–0.129	0.0798	-3.03×10^{-3}	$R^2 = 0.70$; RMSE = 2.458	
50%	2.814	–0.339	–0.0421	0.0614	-2.593×10^{-3}	$R^2 = 0.63$; RMSE = 1.760	
95%	4.840	–0.194	–0.0321	0.0510	-2.214×10^{-3}	$R^2 = 0.59$; RMSE = 1.390	

Of particular interest was our finding that the observed associations between nearshore temperature patterns and site characteristics were similar across lakes. This supports our hypothesis that local (site-specific) factors are the primary drivers of nearshore water temperature variation. Lake-specific differences in lake-wide water temperatures were accounted for with the inclusion of T_{max} terms. Our regression models (Table 1) provide quantitative summaries of the empirical relations that link site and lake-specific variables to seasonal patterns of temperature variation for the Great Lakes.

The importance of site characteristics in shaping nearshore temperature patterns has implications for nearshore productivity. Because temperature is one of the primary determinants of biological productivity, our results suggest that higher productivity levels should be found at shallow, low-fetch sites that warm faster in the spring and remain warmer in summer than other nearshore sites. Furthermore, variation in invertebrate communities (Kilgour et al., 2000) and primary productivity (Steissberg et al., 2005) are related to upwelling occurrence, as are lethal and sublethal effects on fish (Donaldson et al., 2008). Based on our findings, deep, high fetch sites are most likely to experience these ecological effects of upwellings.

Nearshore water temperature projections

Under our chosen climate change scenario, projections of future nearshore surface water temperatures showed that the entire distribution of midsummer water temperatures (i.e. minimum, median and maximum temperatures as characterized by the 5%, 50% and 95% benchmarks) at all sites experienced a progressive shift toward higher values throughout the entire projection period. Midsummer temperatures at our two representative high fetch sites (Kincardine and Marquette) are projected to increase more slowly than lake-wide surface temperatures, whereas temperatures at the two low fetch sites (Sandusky and Kingston) are projected to increase at a similar rate as lake-wide surface temperatures. Therefore, based on our analysis, site characteristics

appear to be important determinants of how nearshore water temperature regimes will be altered by increased surface temperatures and projections of lake-wide mean water temperatures may not accurately characterize future temperatures in nearshore areas. Importantly, the site characteristics we examined (particularly depth and fetch) are easily measured, and therefore our projection approach could be easily applied at other nearshore sites in the Great Lakes. However, it should be recognized that our projection tools summarize gross differences in variability patterns between sites that differ greatly in depth and fetch and are founded on summaries of variability patterns characteristic of historical climatic conditions. More complex, mechanistic projection tools (e.g. Stips et al., 2004) might be needed to deal more precisely with sites that differ in other important physical factors and with climate change scenarios that generate climatic conditions that are very different from those in the historical record.

The differences between the warmest and coolest temperatures experienced at a site are projected to increase slightly because the warmest temperatures at a site (i.e. temperatures associated with the 95% benchmark) will increase more over time than will the coolest temperatures at a site (i.e. those temperatures associated with the 5% benchmark). However, these projected differences are relatively small, e.g. of our four representative sites, the difference between the warmest and coolest temperatures is projected to increase at most by 1.2 °C by 2071–2100 compared to 1971–2000 (for Marquette). This result is dependent on our assumption of a fixed hypolimnetic temperature – if this increases with climate change, then the range of temperatures experienced at a site may remain unchanged or may even shrink.

Our projection models for midsummer nearshore temperatures included fetch terms rather than exposure. In principle, monthly summer exposure values should be more effective than fetch in predicting midsummer nearshore temperatures. This is because exposure values include wind characteristics of a site, and these should be closely associated with upwelling events (Plattner et al., 2006). However, we found fetch-based models to explain more variation in the midsummer

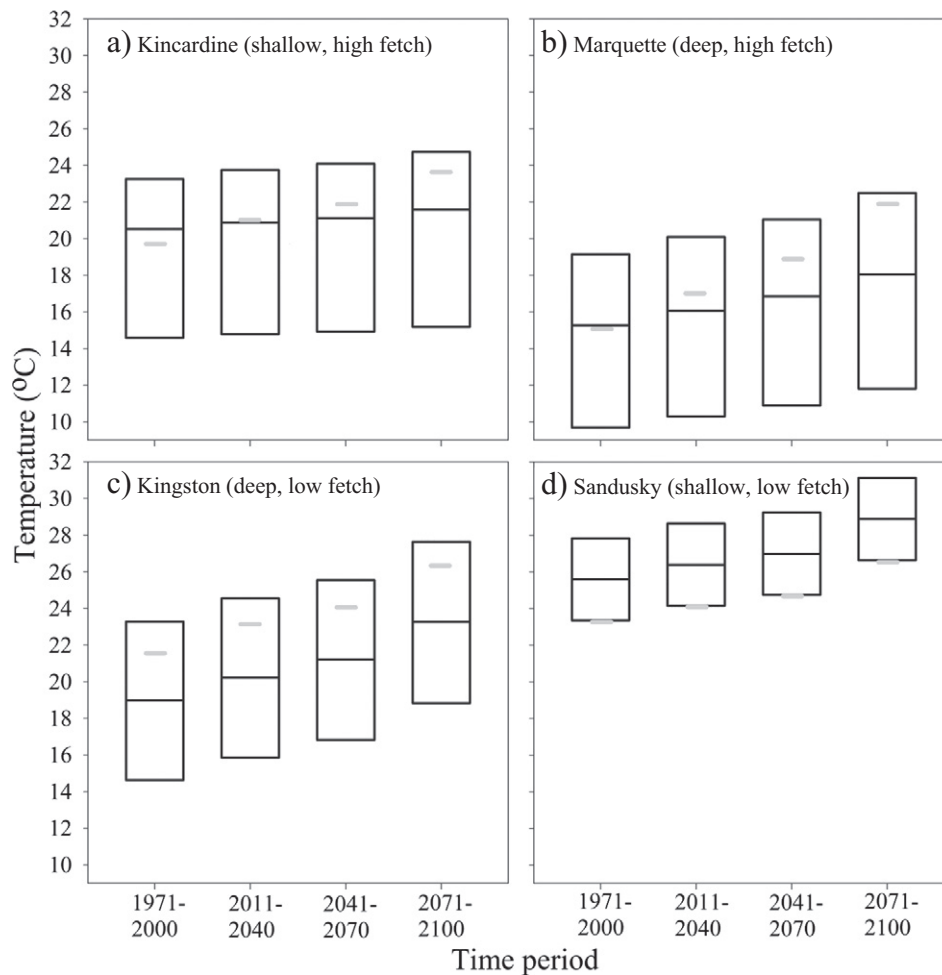


Fig. 5. Projections of the range of midsummer nearshore water temperatures for 1971–2000, 2011–2040, 2041–2070 and 2071–2100 under the A2 CO₂ emission scenario for (a) Kincardine (H3, Lake Huron, shallow and high fetch), (b) Marquette (S1, Lake Superior, deep and high fetch), (c) Kingston (O2, Lake Ontario, deep and low fetch) and (d) Sandusky (E6, Lake Erie, shallow and low fetch). Horizontal lines on the boxes show values of raw nearshore water temperature projections, derived from projected values for the 5% (bottom line), 50% (mid-line) and 95% (top line) benchmarks used to characterize distributions of nearshore–lake-wide water temperature differences. Gray dashes mark the projected lake-wide average surface water temperature.

benchmark values than exposure-based models. A possible explanation for this is the limited wind data available for calculating exposure of Great Lakes sites because historical wind data is restricted to a few onshore sites and time series are often short and sporadic and current wind data is collected offshore but at a very broad spatial scale. This is important because wind data collected at one or, at most, two offshore buoy sites per lake may not accurately reflect conditions at specific nearshore sites. Furthermore, for each site, our exposure values were highly variable across the six years of available offshore wind data. Overall, this suggests that we were unable to capture a meaningful ‘characteristic’ exposure value for our sites, limiting the usefulness of mean monthly exposure as a predictor variable. It is possible that other wind-based predictor variables based on longer time periods or using data more reflective of site-specific conditions could add significant explanatory power to our models. Moreover, projections of the impact of climate change on future wind patterns remain uncertain (Pryor and Barthelmie, 2011); and, until better wind projections are available, fetch-based projection models are likely to be more useful than exposure-based models. Additionally, in large lakes, Kelvin waves generated following thermocline oscillations rotate around the shore of the lake after wind activity has ceased (Beletsky et al., 1997). Because these processes make it likely that upwelling at some sites will be sensitive to distant and past wind conditions, site-specific, immediate exposure values may not be effective in predicting upwelling events. Nevertheless, the midsummer benchmark models selected using AICc-

based criteria contained the same sets of independent variables for each of the benchmarks and had high r^2 values. This suggests that both our fetch and exposure-based models, despite containing predictor variables that were not strictly independent, were able to capture important relationships linking site characteristics to water temperature variation.

Our projections of future nearshore water temperatures were based on one climate change scenario (the IPCC A2 scenario) and therefore should be taken as one possible state of future conditions. Furthermore, the A2 scenario is based on higher population growth and carbon emissions than other IPCC scenarios; therefore, our projections should represent the maximum changes likely for nearshore areas. Importantly, our models could easily incorporate other emission scenarios to produce a wider range of projections. In addition, our projections were based on changes in only one climate-change related variable (lake-wide surface temperatures); however, changes in other variables (e.g. wind and water levels) are likely also important and should be considered in future projection iterations.

Changes in Great Lakes' water temperatures will have important effects on the fish and aquatic habitats of the Great Lakes (Lynch et al., 2010). Our representative shallow, low fetch site, Sandusky, was projected to have the highest midsummer water temperatures and to be warmer overall than the lake-wide surface waters of the lake. Sandusky is also located on Lake Erie, the warmest of the Great Lakes (Trumpickas et al., 2009). Sites such as Sandusky could provide

additional habitat for warm-water fish species which are currently limited in their distribution, growth and/or abundance by cooler than preferred water temperatures (e.g. smallmouth bass *Micropterus dolomieu*; Lynch et al., 2010). Such sites could also be invaded by non-native fish species that are currently excluded from the Great Lakes by low water temperatures, thus establishing source sites in the lakes that could accelerate expansion of such invasions. Furthermore, increases in nearshore water temperatures could exacerbate water quality problems and have negative effects on recreational sites and municipal and industrial water intakes (Schindler, 2001). Intake of Great Lakes waters for human consumptive uses is large with many intake sites and will likely increase in the future. Greater attention to conditions at Ontario's intakes has been mandated since enactment of its Clean Water Act in 2006 with its strong emphasis on source water protection.

Limitations of approach

The goals of this study were to characterize seasonal patterns in nearshore-whole-lake water temperature differences, identify key drivers of nearshore temperature variation on a broad, basin-wide scale and produce a first-generation set of nearshore water temperature projections. Inherent in our approach was a focus on using simple indices, available datasets and straightforward explanatory variables to produce a model that would be readily accessible until more refined and detailed models become available. Such an approach necessarily entails several limitations that can be addressed in future work.

Our models can project the range of temperature values expected at a site over a number of years, but not the temperature for any specific day or year. Furthermore, our models do not attempt to account for variation in water temperatures at a site within a day, which can be substantial (changing up to 10 °C per hour) and have important biological effects (Wells and Parker, 2010). The unavailability of sub-daily temperature data for a long dataset such as ours (1943–2000) precluded this study from addressing within-day water temperature variation.

For the 2071–2100 period, projected $T_{\max}$ values will exceed observed maximum $T_{\max}$ values for all lakes (Trumpickas et al., 2009). Therefore, the nearshore projections which rely on these extrapolated $T_{\max}$ values should be interpreted cautiously.

Our nearshore projections did not consider several climate change related factors that could impact water temperatures in the future. These include: (i) changes in site depth due to changes in Great Lakes' water levels (Angel and Kunkel, 2010), (ii) changes in wind patterns (Pryor and Barthelmie, 2011), (iii) changes in the timing and strength of seasonal phenomena such as ice cover and stratification (McCormick and Fahnenstiel, 1999; Wang et al., 2012), and (iv) changes in hypolimnetic temperatures and thermocline depth (Lehman, 2002). To date, projections for water levels and wind patterns under climate change scenarios remain fraught with uncertainty and further refinement of these projections is needed before these factors could be confidently considered in nearshore temperature projections. However, because depth was included as an explanatory variable in all our models, changes in water levels could be assessed by testing how a range of depth values affect model outputs. Furthermore, the role of internal waves and other hydrodynamic processes were not explicitly considered in our models (Simons and Schertzer, 1987); a mechanistic 3-D simulation model would be needed to deal explicitly with such spatially localized hydrodynamic drivers of local water temperatures.

More detailed, and extensive, wind products exist that could be used to more thoroughly explore the role of wind in shaping nearshore temperature patterns (e.g. National Centers for Environmental Predictions Climate Forecast System Re-analysis). Calculating site-specific wind conditions could be possible with these products, although the process would likely be complex and time-consuming, limiting their applicability to an accessible modeling approach (such as ours) that is intended for use by lake managers.

We expect that the inclusion of these factors in future projections of nearshore water temperatures could considerably improve the accuracy of the projections. Our modeling approach was successful in identifying approaches for future work and provided a limited set of water temperature projections to be considered on a provisional basis until more refined projections are available.

Conclusions

In summary, our results showed that the seasonal pattern of nearshore-lake-wide temperature differences varied greatly from site to site. This seasonal pattern could be characterized using a simple index based on quantile regression of observed site specific temperature data. Differences in the values of this index could be related to easily measured site characteristics. Our generalized approach to linking nearshore water temperatures to easily determined physical characteristics (depth and fetch) of nearshore sites produced simple empirical regression models that could be used for estimating current water temperature regimes at sites that are of interest to managers but where routine temperature monitoring programs are absent. In addition, these models can be used for translating different climate change scenarios into site-specific projections of future water temperature regimes. Such projections can provide the basis for provisional assessments of a wide range of effects that are of concern to managers (e.g. changes in habitat suitability for key life stages of species of management concern, changes in overall biological productivity).

A comparison of our model outputs to independently observed nearshore temperatures would be an important step in validating the results of this study. For example, remote-sensed surface water temperatures for specific nearshore areas in midsummer could be compared against the output of the benchmark models using a zero value for depth and site-specific values for the other independent variables. Furthermore, the models could easily be used to generate projections of midsummer water temperatures under more advanced climate change scenarios, as they become available.

There are several promising avenues for extending and refining our results. For example, lake or sub-lake specific models could be developed to better capture regional variation in water temperature regimes. Such models could have particular applicability in areas of relatively unusual morphology (e.g. the shallow western basin of Lake Erie), areas isolated from the main body of the Great Lakes (e.g. Georgian Bay), and areas of intense management interest (e.g. areas with significant fisheries). Moreover, additional factors such as site-specific wind data, changes in water levels, changes in hypolimnetic temperatures, changes in thermocline depth and changes in the timing and strength of stratification and ice cover could all be considered in future work.

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Appendix A. Electronic supplementary material

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.jglr.2014.11.024>.

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